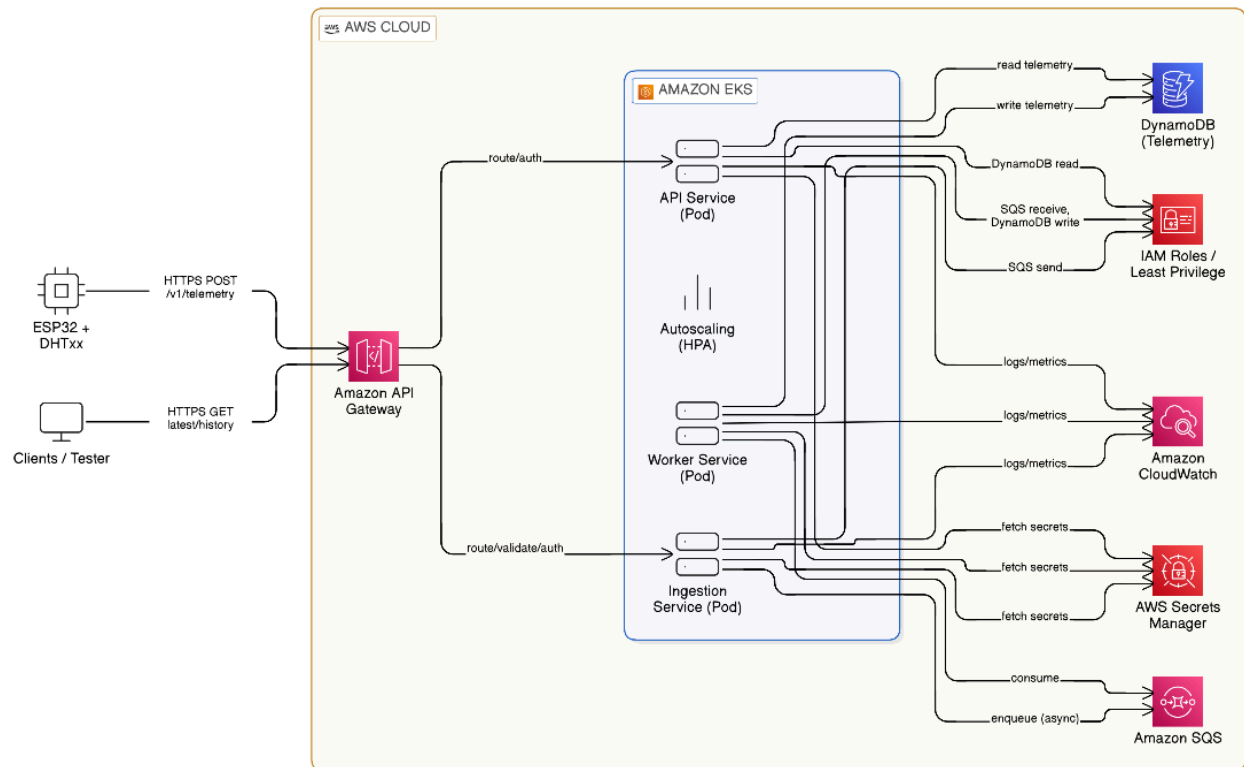


Project Architecture Package
Cloud-Based IoT Environmental Telemetry Platform
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1. Architecture Diagram



2. Components and Responsibilities

Amazon API Gateway (HTTP API)

- TLS termination and routing
- Authentication enforcement (API key or JWT)

Ingestion Service (Amazon EKS)

- Accept telemetry from ESP32
- Validate schema and required fields
- Publish telemetry event to SQS

- Respond quickly

Amazon SQS

- Buffer telemetry events
- Enable async processing and reliability
- Support failure scenarios

Worker Service (Amazon EKS)

- Consume messages from SQS
- Normalize, validate values
- Write to DynamoDB
- Retry on failure, log errors, emit metrics

API Service (Amazon EKS)

- Read telemetry from DynamoDB
- Provide REST endpoints for latest and historical data
- Enforce authorization if needed

3. Data Flow

Flow A: Telemetry ingest (ESP32 to AWS)

1. ESP32 reads DHT sensor
2. ESP32 sends POST /v1/telemetry to API Gateway

3. API Gateway routes to Ingestion Service (Amazon EKS)
4. Ingestion validates and writes message to SQS
5. Worker consumes from SQS and writes to DynamoDB
6. Logs and metrics go to CloudWatch

Flow B: Read latest

1. Client calls GET /v1/devices/{deviceId}/telemetry/latest
2. API Gateway routes to API Service (Amazon EKS)
3. API Service queries DynamoDB
4. Returns JSON

Flow C: Failure demo

- DynamoDB throttling or worker failure → queue depth grows
- CloudWatch alarm triggers on ApproximateNumberOfMessagesVisible
- Recovery: fix worker / scale it → drains queue

4. Key Design Decisions and Trade Offs

- DynamoDB vs. RDS
 - Chosen: DynamoDB
 - Why: serverless scaling, no schema migrations, fits time-series-per-device pattern well
 - Trade-off: complex ad-hoc analytics queries are harder than SQL, you design access patterns up front
- API Gateway and EKS

- Why: clean separation (gateway policies/auth) and Kubernetes required by milestones
- Trade-off: slightly more setup than using ALB-only ingress
- SQS async processing
 - Why: resilience, spike absorption, clean failure demos
 - Trade-off: eventual consistency

5. API Sketch

POST /v1/telemetry (device ingest; returns 202)

GET /v1/devices/{deviceId}/telemetry/latest

GET /v1/devices/{deviceId}/telemetry?start=...&end=...

GET /health